Date: 4 March

M T W T F S S

Lecture 11

Find DTFT of following signal

$$x(n) = \begin{cases} 1 & 0 \le n \le 4 \\ 0 & \text{otherwise} \end{cases}$$

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

$$= \sum_{n=0}^{4} x(n)\, e^{-j\omega n}$$

$$= x(0)e^{-j(0)} + x(1)e^{-j(\omega)} + x(2)e^{-2j\omega} + x(3)e^{-3j\omega}$$

$$= 1 + e^{-j\omega} + e^{-2j\omega} + e^{-3j\omega} + e^{-4j\omega}$$

To find magnitude & phase of signal

Example 4.4.4

$$h(n) = ba^n u(n)$$

$$H(\omega) = \sum_{n=0}^{\infty} ba^n e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} b\left(ae^{-j\omega}\right)^n$$

$$= b \times \frac{\cancel{1-(ae^{-j\omega})^0}}{1-ae^{-j\omega}}$$

$$= \frac{b}{1-ae^{-j\omega}}$$

$$= \frac{b}{1-a(\cos\omega - j\sin\omega)}$$

for minimum value take derivative w.r.t that vari

To plot substitute value in $|X(\omega)|$ ↑ w ki value

for plotting we require minima & maxim

$\tan^{-1}\left(\frac{\text{imag}}{\text{real}}\right)$ for angle

num - den

Date:

$$= \frac{b}{1 + a\cos\omega + jasin\omega}$$

$$|H(\omega)| = \frac{1}{\sqrt{(1-a\cos\omega)^2 + (asin\omega)^2}}$$

$$= \frac{b}{\sqrt{1 - 2a\cos\omega + a^2\cos^2\omega + a^2\sin^2\omega}}$$

$$|H(\omega)| = \frac{b}{\sqrt{1 - 2a\cos\omega + a^2}}$$

$$\frac{d}{d\omega} = \frac{d}{d\omega}\left(1 - 2a\cos\omega + a^2\right)$$

$$= 0 + 2asin\omega + 0$$

$\sin\omega = 0$

$\omega = 0, \pi, 2\pi$

Sampling ... and output signal is same

Date: 6 March

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Lecture

Multirate Digital Signal Processing

In many pratical applications of digital signal processing one is faced with the problem of changing the sampling rate of a signal either increasing it or decreasing it

For example

i) In telecommunication systems that transmit and receive different type of signal (e.g speech, video etc) there is a requirement to process the various signals at different rates according to the corresponding bandwidth of the signals the process of converting a signal from a given rate to a different rate is called sampling rate conversion

In turns systems that employ multiple sampling rates in the processing of digital signal are called digital signal processing signal

ii) Audio Processing

Audio signals can be converted from 44.1 KHz (for CDs) to 48 KHz (DVDs)

iii) Image Processing

Images / Videos can be upscaled to higher

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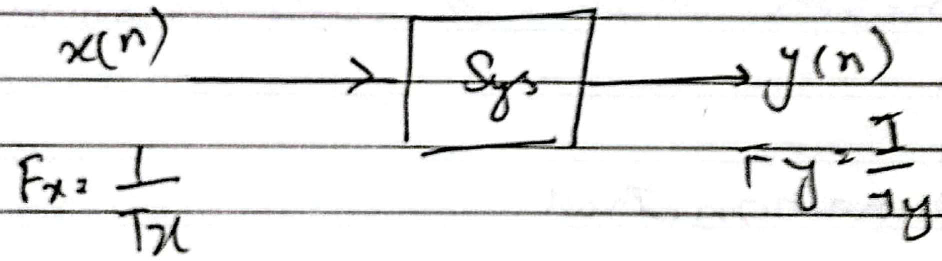
resolution (no of bits increase)

iv) Seismic Data Processing

The size of seismic data can be reduced for storage and transmission (by down sampling & deducing) (any kind of data size can be reduc...

V) Reduced Bandwidth

By reducing the sampling rate of a signal (decimatio... the amount of data that needs to be transmitted / stored is decreased this directly reduces the required bandwidth



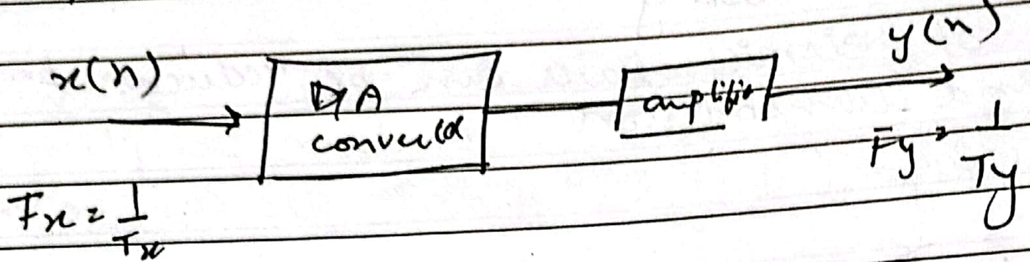
$F_x \neq F_y$

Sampling rate conversion of a digital signal ca... be accomplished in one of ~~the~~ two general method

One method is to pass the digital signal through a D/A converter filter if it necessary and then ~~to~~ resample the resulting analog signal at the desired rate

The second method is to perform the sampling rate conversion entirely in the digital

i) First domain method



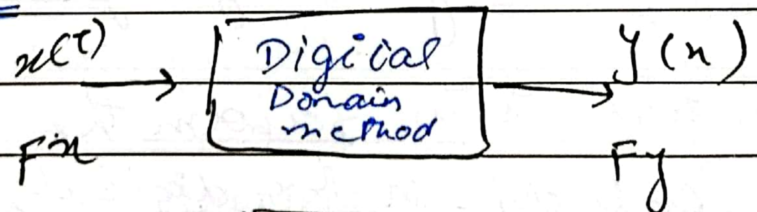
Pros and cons
* The benefit of this method is that Fx and Fy can have any arbitrary value
* The problem in this method is extensive approximation which distorts the signal

ii) Digital Domain method

* In this method sampling rate conversion is ~~permanent~~ performed entirely in the digital domain
* In this method Fx and Fy will have same relationship. However we need NOT to perform D/A converted

$F_y = I F_x$

Up Samples



$F_y > F_x$ | $2 > 3$

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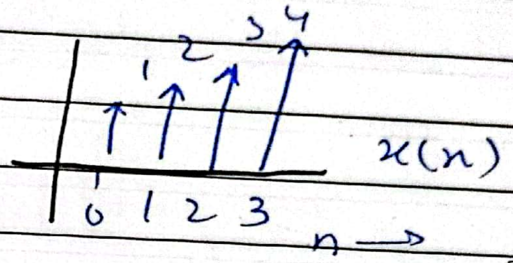
Let

$x(n) = \{1\ 2\ 3\ 4\}$ here 4 samples are taken

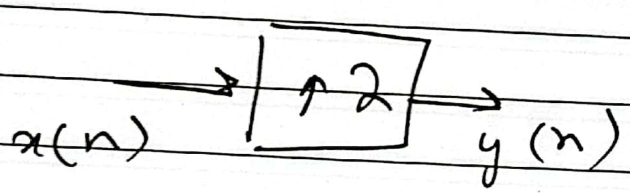
n 0 1 2 3



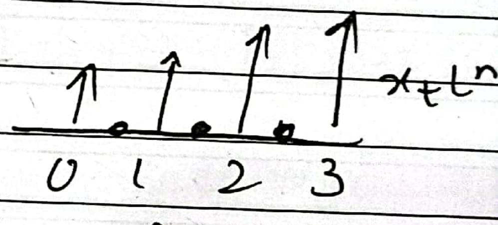
If sampling rate is increased it mean T_x reduces & F_x is increased

here if time is 9 after sampling its time decrease here frequency increase

Let's assume we upsample the signal by a factor of 2



$F_y = 2F_x$



$x_t(n) = \{1, 0, 2, 0, 3, 0, 4\}$

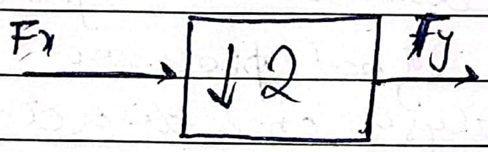
Date: 11 March

Lecture 13

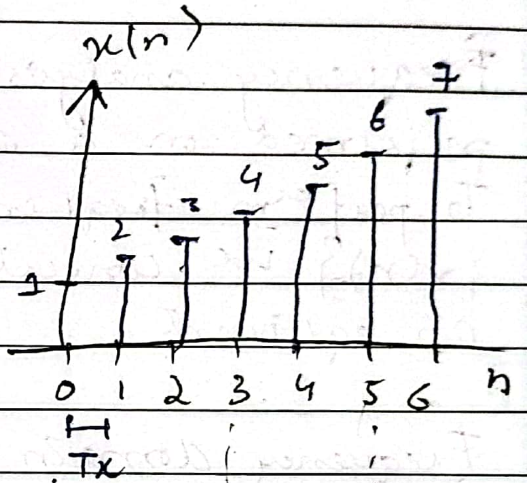
→ input signal sampling frequency is greater than output signal

→ Sampling time is increased

Down Sampler



$F_x > F_y$

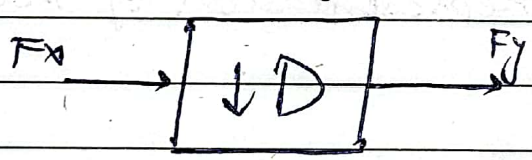


Let $x(n) = \{1, 2, 3, 4, 5, 6, 7\}$

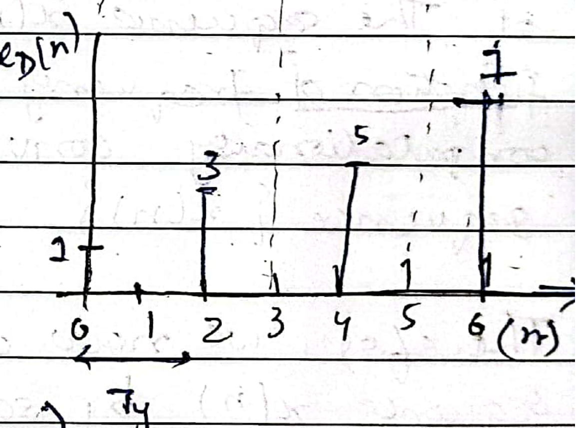
$n = 0, 1, 2, 3, 4, 5, 6$

7 samples converted into

Decimation by factor D



$F_y = \frac{F_x}{D}$



((D-1) samples will be dropped)

→ (2-1) = 1 one sample will be missed

this result if we are firstly sample for 1 sec after that we will sample after every 2 secs

→ (3-1) = 2

2 sample will dropped sampling time will increase cause decrease in frequency

Benefit

Why do we down sample the signal?

→ As No of samples reduces processing become quick (processing time reduce)

→ Half of the sample will store (efficient storage)

KARIMIA NOTES

Chapter 7
Discrete Fourier Transform

Frequency analysis of a discrete time signals is usually performed on a digital signal processor
To perform frequency analysis on a discrete time signal $\{x(n)\}$ we convert the time domain sequence to an ~~equivient~~ equivalent

Frequency domain representation we know that such a representation is given by the fourier transform $X(\omega)$ of the sequence $x(n)$ - However, $X(\omega)$ is continuous function of frequency and therefore it is not a computationally convienient representation of the sequence $\{x(n)\}$

Therefore, we now consider the representation of a sequence $x(n)$ by samples of its spectrum $X(\omega)$. Such representation is called Discrete Fourier Transform (DFT)

If continuous → discrete time fourier transform

If discrease → discrete fourier

Date:

Topic 7.1 Frequency domain sampling : The Discrete fourier transform

7.1.1 Frequency domain sampling of Discrete Time signals

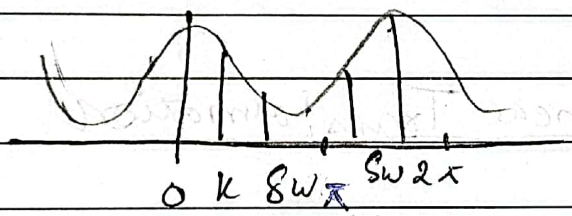
Recall that

$$X(w) = \sum_{n=-\infty}^{\infty} x(n) e^{-jwn} \rightarrow 7.1.1$$

↳ Here it is continuous.

anything that is continuous cannot apply on processor, we need to convert it into fourier transform

$X(w)$



fourier Transform is periodic at 2π

Suppose that we sample $X(w)$ periodically in frequency at spacing of δw radians between successive samples where $X(w)$ is periodic with period 2π

↳ space b/w two samples

Total samples =

⇒ For convenience, we take N equidistant samples in the interval $0 \le w < 2\pi$ with spacing $\boxed{\delta w = \frac{2\pi}{N}}$ as shown in figure → in 1 period

⇒ If we evaluate $(7.1.1)$ at $w = \frac{2\pi k}{N}$ we obtain → Now it is equidistant sample

$$X\left(\frac{2\pi k}{N}\right) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\frac{2\pi}{N}kn} \qquad k = 0, 1, \ldots N-1$$

↳ inplace of w

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Lecture 14

Discrete Fourier Transform

w is replace

$K = 0, 1 \ldots N-1$

DFT (focus) $X(K) = \sum_{n=0}^{N-1} x(n) \, e^{-j\frac{2\pi kn}{N}} \longrightarrow (1)$

$n = 0, 1, 2 \ldots N-1$

IDFT $\underbrace{x(n)}_{\text{output}} = \frac{1}{N} \sum_{k=0}^{N-1} \underbrace{X(k)}_{\text{input}} e^{j\frac{2\pi kn}{N}} \longrightarrow (2)$

→ we intend to calculate 4 point DFT n point
n can be something

The DFT as a Linear Transformation

The formulas in eq (i) and eq (ii) may be expressed as

Observe difference among both

(Focus) $X(K) = \sum_{n=0}^{N-1} x(n) \, W_N^{kn}$ $\quad K = 0, 1, \ldots N-1$

compare with 2 $\boxed{x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(K) \, W_N^{-kn}}$ $\quad n = 0, 1, \ldots N-1$

where

difference b/w both equation $\boxed{W_N = e^{-j\frac{2\pi}{N}}}$

4 Point DFT

↳ n can be n $\boxed{N = 4}$ $\quad K = n = 0 : 3$

For

0 to 3
As N-1
4-1=3

$K = 0$

W^{kn} → here both ~~K & n are zero~~

$x(0) W_4^0 + x(1) W_4^0 + x(2) W_4^0 + x(3) W_4^0$

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$K = 1$ $\rightarrow$ b/c $N = 4$ $k, \ln, 1$
$n = 0 \; K = 1$

$x(0) W_4^0 + x(1) W_4^1 + x(2) W_4^2 + x(3) W_4^3$

b/c of n 4 term as n is from 0 to 4

$K = 2$

$x(0) W_4^0 + x(1) W_4^2 + x(2) W_4^2 + x(3) W_4^3$

$K = 3$

$x(0) W_4^0 + x(1) W_4^3 + x(2) W_4^6 + x(3) W_4^9$

$\Rightarrow W_N x_N = X(K)$
$\downarrow$
4 point DFT

$$W_N = \begin{bmatrix} W_4^0 & W_4^0 & W_4^0 & W_4^0 \\ W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ W_4^0 & W_4^2 & W_4^4 & W_4^6 \\ W_4^0 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix}_{4\times 4}$$

x is vector
W is matric

$\downarrow$
This is W_N matric ~~This would~~ be multiply with x_N and give $X(K)$

$$W_N x_N = X(K)$$
$4\times4 \quad 4\times1 \quad 4\times1$
$\downarrow$
4 point DFT

Size of matric is N depend
if $N = 5$ then 5×5
if $N = 2$
2×2

Calculate $W_4^0, W_4^1, W_4^2, W_4^3 = ?$

for W_4^0

$$W_4^0 = W_4^{kn} = e^{-j\frac{2\pi(0)}{4}} = e^0 = 1$$

for W_4^1

$$W_4^1 = W_4^{kn} = e^{-j\frac{(2\pi)(1)}{4}} = e^{-j\frac{\pi}{2}} = \cos\frac{\pi}{2} - j\sin\frac{\pi}{2} = 0 - j(1) = -j$$

for W_4^2

$$W_4^2 = W_4^{kn} = e^{-j\frac{(2\pi)(2)}{4}} = e^{-j\pi} = \cos\pi - j\sin\pi = -1 - 0 = -1$$

for W_4^3

$$W_4^3 = W_4^{kn} = e^{-j\frac{(2\pi)(3)}{4}} = e^{-j\frac{3\pi}{2}}$$
$$= \cos\left(\frac{3\pi}{2}\right) - j\sin\left(\frac{3\pi}{2}\right)$$
$$= 0 - j(-1) = j$$

KARIMIA NOTE

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Periodicity

$$W_N^k = W_N^{k+N}$$

By periodicity property

$$W_4^0 = W_4^4$$

$$W_4^1 = W_4^5 = W_4^9$$

$$W_4^2 = W_4^6$$

So

$$W_N = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & j \end{bmatrix}$$

Example 5.1.3 Compute the DFT of the 4 point sequence

$$x(n) = [0 \ 1 \ 2 \ 3]_{1\times4}$$

$$W_N x_n = x(k)$$

$$\underset{4\times4}{\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}} \underset{4\times1}{\begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}} = \begin{bmatrix} 6 \\ -2+2j \\ -2 \\ -2-2j \end{bmatrix} = x(k)$$

to make $x(n)$ size same to W_N we will do zero padding

$x(n) = [0 \ 1]$

$x(n) = [0 \ 1 \ 0 \ 0]$

it make size equal also remove aliasing effect

Lecture 15

No. of blocks = l

No of element / Summation in each block = N

$L \rightarrow$ points in original $x(n)$

If we replace $n = n - lN$

$n = lN - lN = 0$ (Lower)

$n = \cancel{lN} + N - 1 - \cancel{lN} = N+1$ (upper)

$x\left(\frac{2\pi}{N}k\right) =$

$x_p \rightarrow$ periodic

$x(n) \rightarrow$ aperiodic

$\rightarrow$ no aliasing

$\Rightarrow$ If $N > L$ than you can reconstruct $x(n)$

$N \downarrow$ Length of $x_p(n)$

$L \downarrow$ Length of original signal

$\rightarrow$ aliasing

$\Rightarrow$ If $N < L$ (some samples remain but signal start repeating itself that result in aliasing)

(remove some point of itself x_p)

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$N \geq L \quad x(n) = x_p(n)$

$0 \leq n \leq N-1$

Topic 5.1.2

Zero padding is done just for better display of fourier transform

→ addition of zeros

L
↓
No of point of input

$$X(\omega) = \sum_{n=0}^{L-1} x(n) e^{-j\omega n} \quad 0 \leq \omega \leq 2\pi$$

exp: L = 10
N should be greater than L

for convenience $L-1$ is replace by $N-1$

DTFT → continous

$xp(n) \rightarrow$ periodic

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Lecture 16

Circular Symmetries of a Sequence

As we have seen the N point DFT of a sequence $x(n)$ of length $N \geq L$ is equivalent to the N point DFT of a periodic sequence $xp(n)$ of period N which is obtained by periodically extending $x(n)$ that is

$$xp(n) = \sum_{l=-\infty}^{\infty} x(n - lN)$$

repeation of $x(n)$

this is shifted by $n-k$

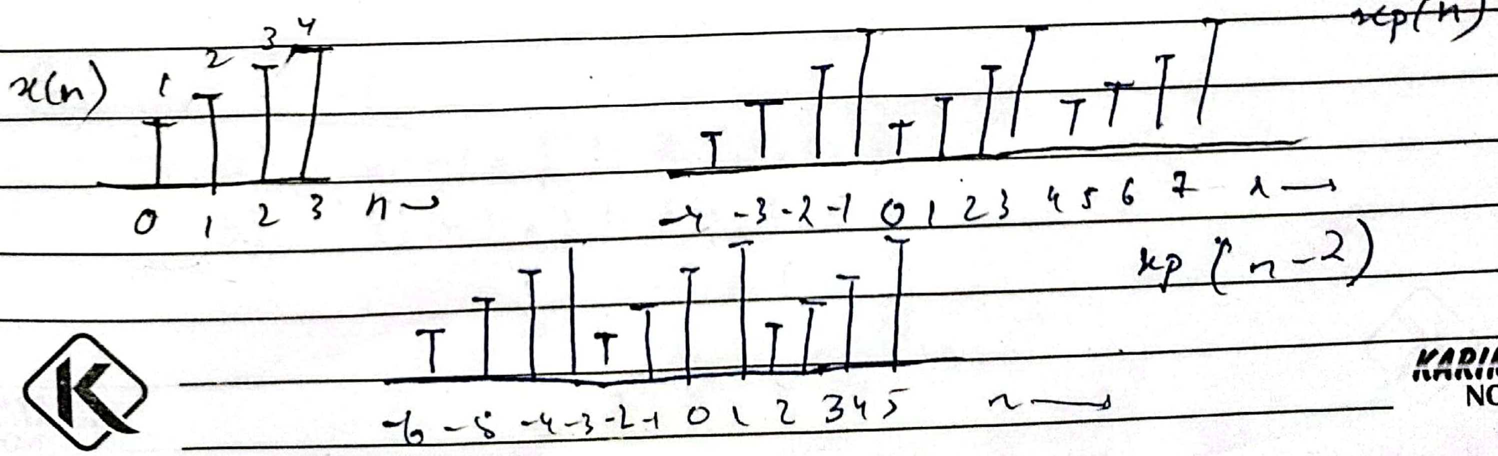
Now suppose we shift the periodic sequence $xp(n)$ by 'k units' to the right, thus we obtain another periodic

$$x'p(n) = xp(n-k) = \sum_{l=-\infty}^{\infty} x(n-k-lN)$$

The finite duration sequence

$$x'(n) = \begin{cases} xp'(n) & 0 \leq n \leq N-1 \\ 0 & \text{otherwise} \end{cases}$$

is related to the $x(n)$ by a circular shift as shown in fig 1 for $N=4$.



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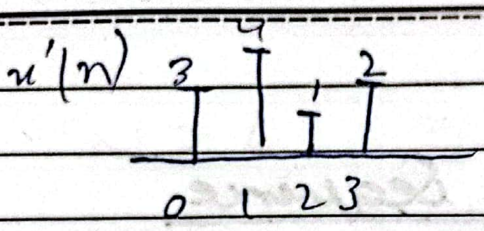

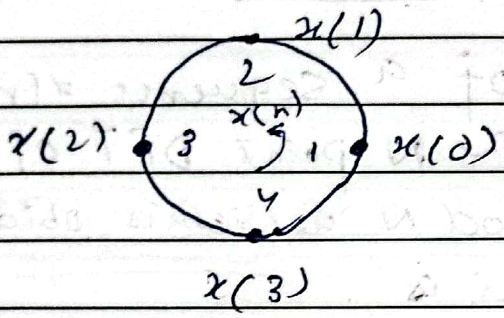

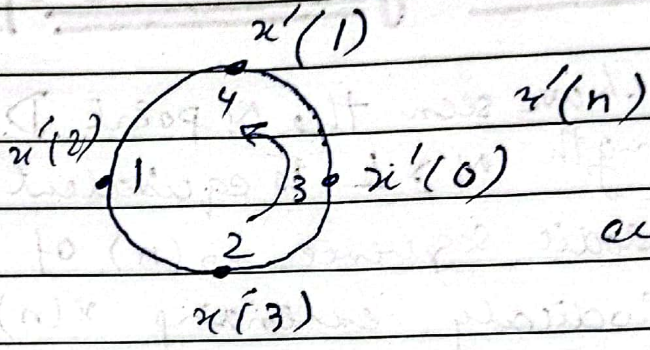

circular shift

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Lecture 17

Chapter 3

Laplace transform
↓
Time domain

Z transform
↓
frequency domain

Pole & zero are used to analyze whether the signal is stable or not

Topic 3.1

3.1.1 Direct Z transform

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

where z is a complex variable

transform time domain in frequency domain

$$X(z) = Z\{x(n)\}$$

$$x(n) \xleftrightarrow{Z} X(z)$$

Z transform infinite power series exist for only values of z where series converges
↓
ROC (region of convergence) attain finite value

Example 3.1.1

a) $x_1(n) = \{1, 2, 5, 7, 0, 1\}$

b) $x_2(n) = \{1, 2, \underset{\uparrow}{5}, 7, 0, 1\}$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

KARIMIA NOTES

CamScanner

Date: M T W T F S S

$$x(z) = x(0)z^{-0} + x(1)z^{-1} + x(2)z^{-2} + x(3)z^{-3} + x(4)z^{-4} + x(5)z^{-5}$$

$$X(z) = 1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + 0 + 1z^{-5}$$

$$X(z) = 1 + \frac{2}{z} + \frac{5}{z^2} + \frac{7}{z^3} + \frac{1}{z^5}$$

not exist at $z=0$ bcz become infinity

b) $x_2(n) = \{1, 2, \underset{\uparrow}{5}, 7, 0, 1\}$

$$= x(-2)z^{2} + x(-1)z^{1} + x(0)z^{0} + x(1)z^{-1} + x(2)z^{-2} + x(3)$$

$$X(z) = 1z^2 + 2z + 5 + 7z^{-1} + 0 + z^{-3}$$

$$X(z) = 1z^2 + 2z + 5 + \frac{7}{z^{1}} + \frac{1}{z^3}$$

doesnot exist at $z=0$ and $z=\infty$
↳ here series diverge to infinity

Characteristics

we can express the sum of finite or Infinite we can express them in close form

Example 3.12 $\quad x(n) = \left(\frac{1}{2}\right)^n u(n)$

$$x(n) = \left\{1, \left(\frac{1}{2}\right), \left(\frac{1}{2}\right)^2, \left(\frac{1}{2}\right)^3, \ldots \left(\frac{1}{2}\right)^n\right\}$$

Z transform

$$X(z) = 1 + \frac{1}{2}z^{-1} + \left(\frac{1}{2}\right)^2 z^{-2} + \left(\frac{1}{2}\right)^n z^{-n} \ldots$$

$$\sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n z^{-n} = \sum_{n=0}^{\infty} \left(\frac{1}{2}z^{-1}\right)^n$$

bcz unit step

KARIMIA NOTES

$1 + A + A^2 + A^3 \dots = \frac{1}{1-A}$ $|A| > 1$ main

$X(z) = \frac{1}{1 - \frac{1}{2} z^{-1}}$. ROC $|z| > \frac{1}{2}$

Example 3.1.3

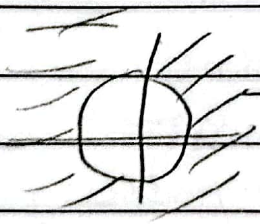
$$x(n) = a^n u(n) = \begin{cases} a^n & n \ge 0 \\ 0 & n < 0 \end{cases}$$

$$X(z) = \sum_{n=0}^{\infty} x(n) z^{-n}$$

$$= \sum_{n=0}^{\infty} (az^{-1})^n$$

$|az^{-1}| < 1$ $|z| > |a|$

$X(z) = \frac{1}{1 - az^{-1}}$ if $z = a$ than ∞ it will not exist



ROC

can take less than and equal to a because it will not exist

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Example 3.1.4

$$x(n) = -a^n u(-n-1)$$

$$X(z) = \sum_{n=-\infty}^{-1} (-a^n) z^{-n}$$

$$= \sum_{l=1}^{\infty} (a^{-1}z)^l$$

$$A + A^2 + A^3 + A^4 = A(1 + A + A^2 + \ldots) = \frac{A}{1-A}$$

this is important

here

$$\frac{a^{-1}z}{1-a^{-1}z} = \frac{1}{1-az^{-1}}$$ (here this form is not valid)

ROC

$|z| < |a|$

↓

We have to see behaviour of power series

⇓

See original expression of $X(z)$

$$\frac{1}{1-A} \rightarrow$$ closed form expression

infinite to finite by changing

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Lecture

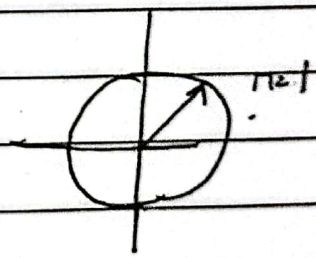
$Z[x(n)] = x(Z) \rightarrow$ complex plane

$$\sum_{n=-\infty}^{\infty} x(n) Z^{-n} = X(Z)$$

$$Z = re^{j\omega}$$
$$= r(\cos\omega + j\sin\omega)$$
$$= \underbrace{r\cos\omega}_{\text{Real}} + \underbrace{rj\sin\omega}_{\text{imaginary}}$$

if $r = 1$ than z transform is equal to fourier transform

$$\sum_{n} x(n)(re^{j\omega})^{-n} \qquad r = 1$$



$$\sum_{n} x(n)\, e^{j\omega n}$$